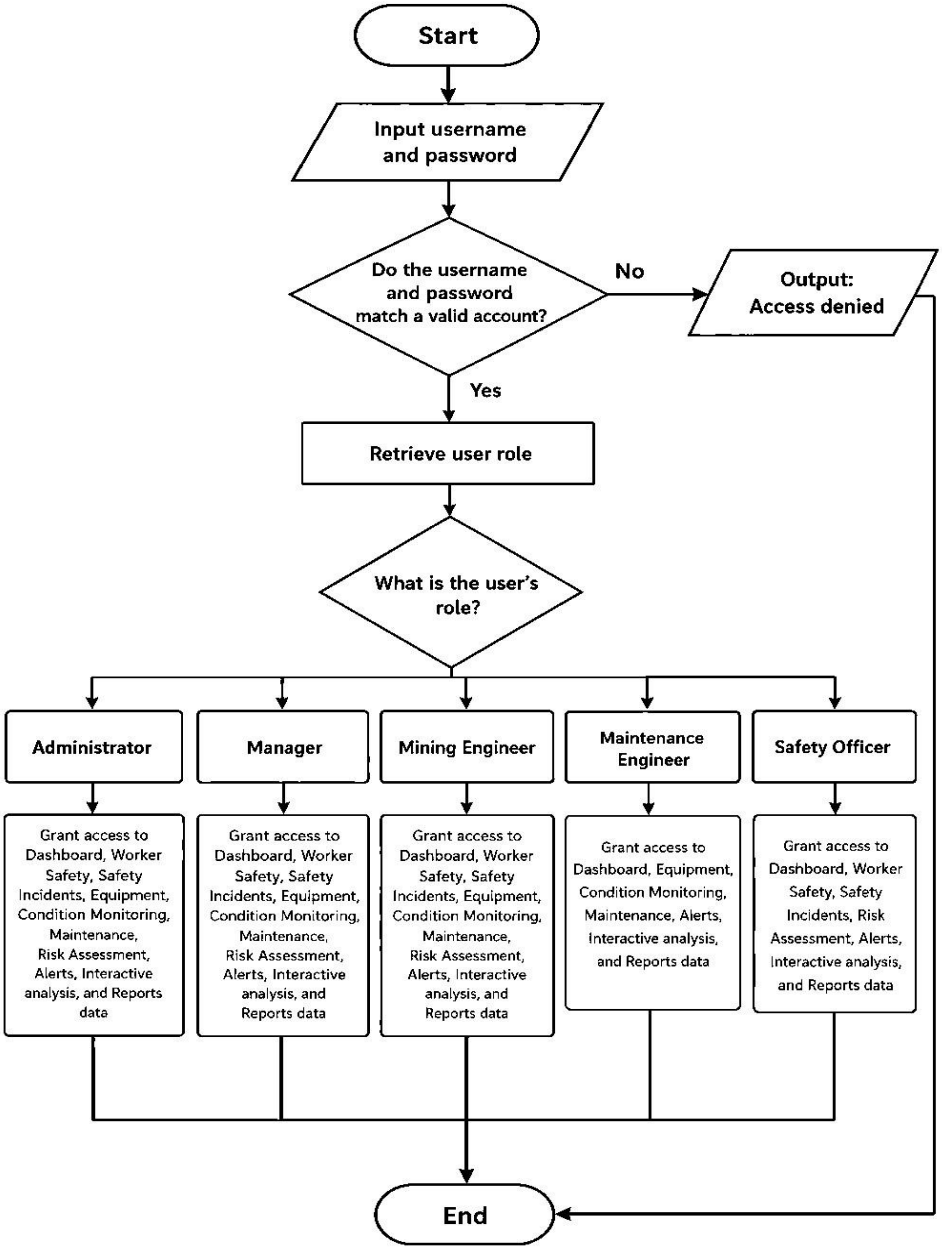
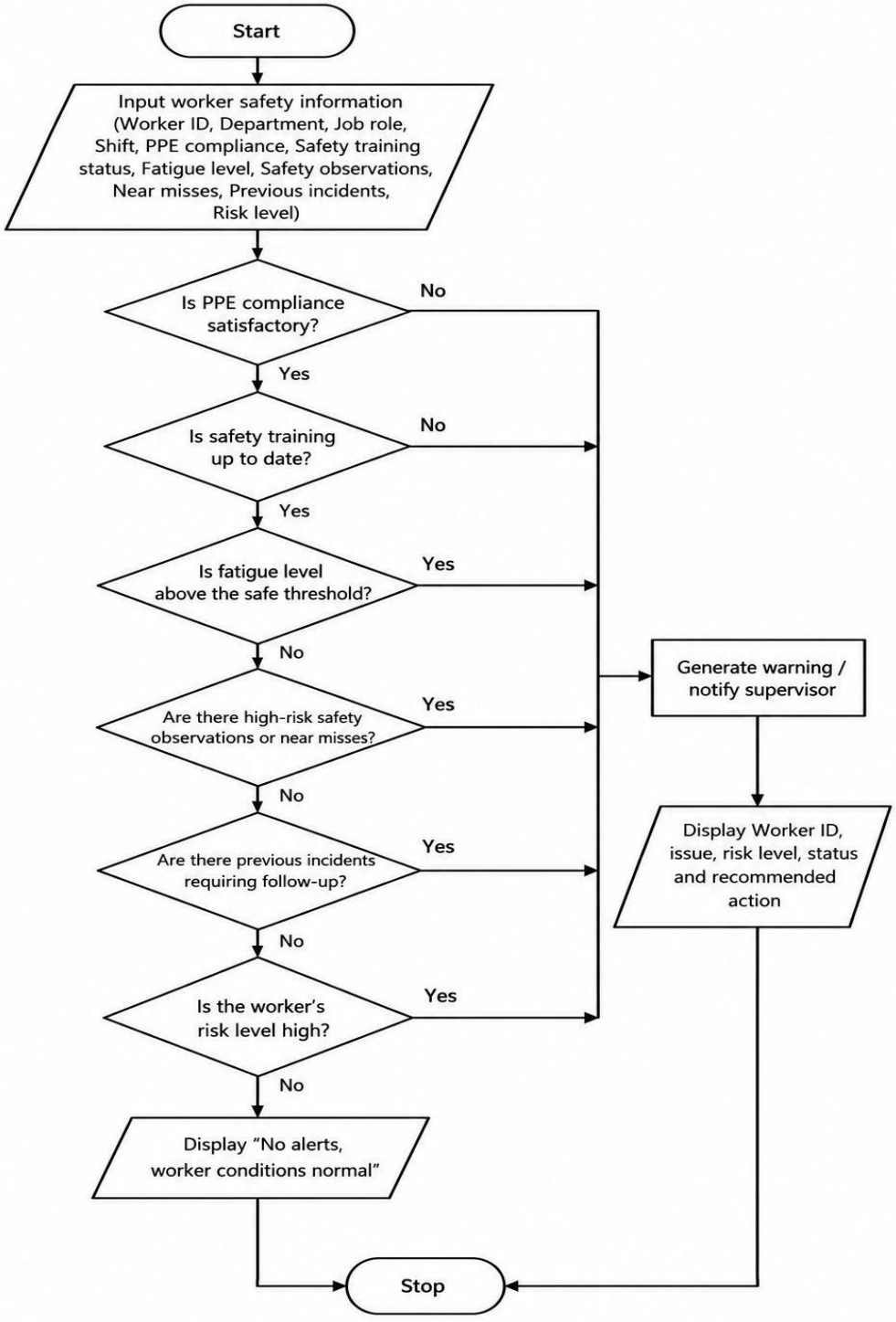
PART 1 FLOWCHART



PARTB


Start
Input worker safety information (Worker ID, Department, Job role, Shift, PPE compliance, Safety training status, Fatigue level, Safety observations, Near misses, Previous incidents, Risk level)
Is PPE compliance satisfactory?
No
Yes
Is safety training up to date?
No
Yes
Is fatigue level above the safe threshold?
Yes
No
Are there high-risk safety observations or near misses?
Yes
No
Are there previous incidents requiring follow-up?
Yes
No
Is the worker's risk level high?
Yes
No
Display "No alerts, worker conditions normal"
Generate warning / notify supervisor
Display Worker ID, issue, risk level, status and recommended action
Stop

PSEUDOCODE

PART A

```
IMPORT data from excel
DEFINE permissions to check login details
Create variable username and password
INPUT username from dataset
INPUT password from dataset
CREATE variable user which will check username where on data where condition is true  IF
user.empty:
      Display “(username not found”)  ELSE:
       Display (“username found”)
CREATE variable stored_pword  IF
password == stored_pword:
   Display ("Login successful")
   CREATE variable role
   DISPLAY (“welcome”, username,” your role is”, role)  IF
role in permissions:            allowed_functions =
permissions[role]
          Display ("You can access:")
          FOR function in allowed_functions:
                Display ("-", function)
CREATE variable choice
INPUT function you want to access on choice
if choice in allowed_functions:            Display
("Access granted to", choice)         else:
          Display ("Access denied")  else:
DISPLAY ("Unknown role")  if
role == "Admnistrator":
DISPLAY ("Full acces granted")
elif role == "Safety Officer":
DISPLAY ("Access to safety functions granted")  elif
role == "Mining Engineer":
DISPLAY ("Access to mining engineering functions granted")  elif
role == "Maintenance Engineer":
DISPLAY ("Access to maintenance functions granted")  elif
role == "Manager":
DISPLAY ("Access to management functions granted")  else:
DISPLAY ("Unknown role")  else:
DISPLAY ("Incorrect password")
```

PART B

START
LOAD the data Worker Safety from excel.
SHOW the page of Worker Health and Safety.
SHOW the records of all worker safety.
User is allowed to search for a worker using Worker ID.

IF the user selects "Add Worker" THEN
INPUT Worker ID.
    INPUT Department.
    INPUT PPE Compliance.
    INPUT Safety Training Status.
    INPUT Fatigue Level.
    INPUT Risk Level.

     Save the new worker record.
        SHOW "Worker added successfully".
     ELSE
        Display an error message.
END IF
ALLOW user to check the safety conditions of the workers.

IF PPE Compliance is "No" OR
     Safety Training is "Incomplete" OR
     Fatigue Level is "High" OR
     Risk Level is "High" or "Critical" THEN

DISPLAY a safety warning.

ELSE
     Show "No safety warnings detected".
END IF
Allow the user to have the access of downloading worker records.
END